

2025 CpSc H Q6

Section: Computer Systems

Topic: Data Representation — Vector vs Bit-mapped Graphics

Question Summary

Two scaled graphics are shown (Graphic A and Graphic B). (a) Identify which graphic is the vector and justify; (b) state one advantage of using bit-mapped graphics compared to vector graphics.

Worked Solution

(a) A vector graphic is defined by shapes and equations (lines, curves, fills). When scaled, edges remain perfectly smooth with no pixelation. A bit-mapped (raster) image is made from pixels; when scaled up, individual pixels become visible (jagged/blurry edges).

In the exam figure, the vector image is the one that still looks crisp after scaling (no stair-steps along edges). Therefore: Graphic A is the vector (identified by its smooth, non-pixelated edges when enlarged).

(b) One advantage of bit-mapped graphics: they are better for photographic/very detailed images with complex colour gradations and textures; you can also edit them at the pixel level.

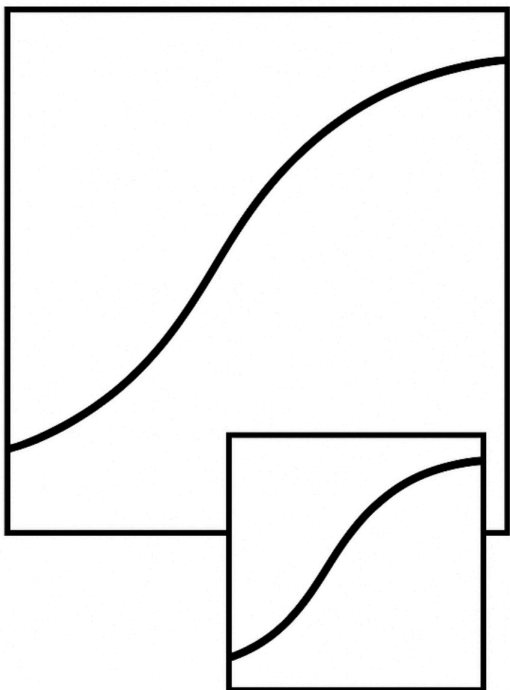
Final Answer

(a) Graphic A — because its edges remain smooth with no pixelation when scaled (vector property). (b) Bit-mapped graphics are superior for realistic photos/complex detail and allow per-pixel editing.

Revision Tips

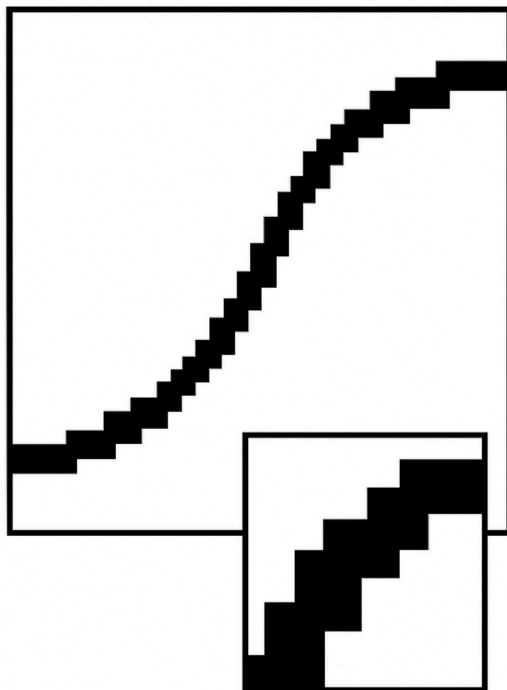
- Vector = defined by maths; scales up or down without losing quality. Raster = grid of pixels; enlarging reveals pixels.
- Files: vectors tend to be smaller for simple shapes; rasters can be large for high resolution.
- Use vectors for logos/icons/line art; rasters for photos and complex textures.

Vector



Smooth

Bitmap



Pixelated